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The first part is based on the algebraic curve in complex field.

# Chapter 1 Fundamental Concepts

## 1.1 The Plane Curve in Projective Plane $\mathbb{P}^2\mathbb{C}$

The knowledge is collected here: ??

## 1.2 Riemann Surface

The knowledge is collected here: ??

## 1.3 Holomorphic and Meromorphic Functions

## 1.4 Holomorphic and Meromorphic Differentials

## 1.5 Differential Forms

## 1.6 Poincare-Hopf Theorem

## 1.7 Complex Manifolds

## 1.8 Algebraic Varieties

## 1.9 Smooth Points, Tangent Spaces and Implicit Function Theorem

## 1.10 Holomorphic Mapping from Compact Riemann Surface into Complex Projective Space

# Chapter 2    The Normalization Theorem

## 2.1   Singularities of Plane Algebraic Curves

Given here:

## 2.2   Connectedness

## 2.3   Normalization

# Chapter 3   The Riemann-Roch Theorem